lj_calculations

May 1, 2026

1 Light Jet Premium Extraction Audit — Phenom 300 Fleet

Segment: Light Jet | **Fleet:** Phenom 300 | **Data pipeline:** Strategic Rhombus — `final_wingx_24_25_lj` reference notebook

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1.1 Purpose

This notebook identifies the **highest-value pricing windows** across 20 LJ strategic corridors. A corridor qualifies only when both signals are simultaneously present in the same time window:

Criterion	Threshold	Meaning
Strong market demand	TOD velocity 15%	Window captures 15% of all corridor flights
Meaningful WUP presence	WUP share 2.0%	WUP holds a real position in that window

The output is an actionable **Premium Extraction Matrix**: which corridors, which windows, and how hard to push on price.

1.2 Data Sources

File	Records	Coverage
`lj_dow_heatmaps.json`	140	20 corridors × 7 days-of-week
`lj_tod_heatmaps.json`	120	20 corridors × 6 time-of-day buckets

Pre-computed corridor aggregates from the Strategic Rhombus pipeline. Raw flight-level data is processed upstream — this notebook ingests summary records only.

1.3 Column Glossary

Column	Definition
`tod_market_velocity_pct`	% of corridor's annual flights occurring in this TOD window — measures market depth
`tod_wup_share_pct`	WUP flights ÷ total flights in that window — the core extraction signal
`dow_market_velocity_pct`	% of corridor's annual flights on this day-of-week
`dow_wup_share_pct`	WUP flights ÷ total flights on that day-of-week

1.4 Analysis Pipeline

Cell	Output	Purpose
1–3	`df_dow_final`, `df_tod_final`	Load and normalize raw heatmap records
4	`df_best` (20 rows)	Rank-sum selection — one best DOW day + one best TOD slot per corridor
5	`df_categorized` (20 rows)	Premium extraction classification into 4 tiers
6	`df_value` (20 rows)	Revenue opportunity quantification per corridor
7–8	`df_signals`	Active corridors only (No Signal removed)

```
Loading heatmap data…
  DOW records : 140
  TOD records : 120
```

```
DOW shape : (140, 4)
Corridors : 20
Days      : ['Friday', 'Monday', 'Saturday', 'Sunday', 'Thursday', 'Tuesday',
'Wednesday']
```

```
                         corridor  dimension  dow_market_velocity_pct  \
0  South Florida  New York City     Monday                  17.0013
1  South Florida  New York City    Tuesday                  14.3329
2  South Florida  New York City  Wednesday                  13.1385
3  South Florida  New York City   Thursday                  12.5540
4  South Florida  New York City     Friday                  13.2147
5  South Florida  New York City   Saturday                  11.6137
6  South Florida  New York City     Sunday                  18.1449

   dow_wup_share_pct
```

```
0            4.6338
1            4.4326
2            4.2553
3            4.6559
4            2.6923
5            3.5011
6            2.6611

TOD shape : (120, 4)
Corridors : 20
Buckets   : ['07:00-09:59', '10:00-12:59', '13:00-15:59', '16:00-18:59',
'19:00-21:59', 'Off-Peak (Late/Early)']

                        corridor             dimension  \
0  South Florida  New York City            07:00-09:59
1  South Florida  New York City            10:00-12:59
2  South Florida  New York City            13:00-15:59
3  South Florida  New York City            16:00-18:59
4  South Florida  New York City            19:00-21:59
5  South Florida  New York City  Off-Peak (Late/Early)

   tod_market_velocity_pct  tod_wup_share_pct
0                  21.7535             4.5561
1                  30.9022             3.7007
2                  24.1169             4.5311
3                  14.2313             3.3929
4                   4.8793             2.0833
5                   4.1169             0.0000
```

1.5 Data Structure After Normalization

Each JSON file contains one record per corridor-dimension pair. Example TOD record:

```
{ "corridor": "DMV  South Florida",
  "dimension": "10:00-12:59",
  "tod_market_velocity_pct": 31.30,
  "tod_wup_share_pct": 11.59 }
```

Two critical properties of this data:

1. **Velocity and WUP share are frequently anti-correlated on the same corridor.** The busiest time window is often *not* the one where WUP is most concentrated. A naive "pick the max velocity slot" approach would systematically select the wrong window. The rank-sum algorithm (Cell 4) resolves this by jointly optimizing both signals.

2. **`tod_wup_share_pct` is WUP's share *within the window*, not of total corridor volume.** A value of 11.59% on DMV → South Florida at 10:00–12:59 means roughly 1-in-9 flights in that slot is WUP — a very strong concentration. Summing velocity across all slots will exceed 100%; they are conditional shares, not a partition.

1.6 Rank-Sum Dimension Selection — How It Works

For each corridor we pick **one DOW day** and **one TOD slot** where velocity and WUP share are simultaneously highest. A simple "pick the max" approach fails because the fastest slot is rarely the one with the highest share. Instead we use a **rank-sum**: rank each slot independently on both metrics, add the ranks, and pick the slot with the lowest total.

1.6.1 Algorithm (applied separately to DOW and TOD)

1. For every slot in a corridor, assign `vel_rank` — rank **1 = fastest** (descending).
2. For every slot in a corridor, assign `wup_rank` — rank **1 = highest share** (descending).
3. `combined_rank = vel_rank + wup_rank`
4. Pick the slot with the **lowest** `combined_rank`.
5. Tiebreaker (equal `combined_rank`): highest `wup_share`, then highest `velocity`.

A slot that ranks #1 on both gets `combined_rank = 2` — the theoretical minimum with any number of slots.

1.6.2 Hand-Trace Example — DMV → South Florida (TOD)

Time Slot	tod_velocity	tod_wup	vel_rank	wup_rank	combined_rank
07:00–09:59	21.1%	9.3%	3	4	**7**
10:00–12:59	**31.3%**	**11.6%**	**1**	**1**	**2** ← winner
13:00–15:59	24.5%	10.1%	2	2	**4**
16:00–18:59	15.3%	9.4%	4	3	**7**
19:00–21:59	4.9%	6.4%	5	5	**10**
Off-Peak	2.9%	4.7%	6	6	**12**

`10:00-12:59` ranks **#1 on velocity AND #1 on WUP share** → `combined_rank = 2` → selected.

1.6.3 What the ranks protect against

Scenario	Pure-velocity pick	Pure-wup pick	Rank-sum pick
Fastest slot has near-zero share	wrong slot	—	balanced
Best-share slot is off-peak (vel 2%)	—	wrong slot	balanced
Both metrics peak at same slot	correct	correct	correct

North Florida → Dallas edge case: all 6 TOD slots have perfectly anti-correlated velocity and WUP (fastest = lowest share). Every slot scores `combined_rank = 7`. Tiebreaker picks Off-Peak (highest wup = 2.63%), which correctly lands in **No Signal** — no actionable premium window exists on this corridor.

```
Best-dimension table: (20, 7)  (expect 20 x 7)
```

	corridor	dow_dimension	tod_dimension	dow_velocity_pct	tod_velocity_pct	dow_wup_share_pct	tod_wup_share_pct
0	Chicago Denver	Monday	10:00-12:59	14.5533	31.3305	3.2172	2.8643
1	Chicago South Florida	Friday	13:00-15:59	16.3783	23.1257	3.3451	3.9900
2	DMV South Florida	Friday	10:00-12:59	17.0175	31.3032	9.4737	11.5880
3	Dallas Denver	Friday	10:00-12:59	15.8143	35.9166	3.7313	3.0668
4	Dallas Phoenix Valley	Sunday	13:00-15:59	15.0358	29.5943	2.7778	3.4274
5	Dallas South Florida	Wednesday	10:00-12:59	16.2885	27.3828	3.0960	3.6832
6	Detroit South Florida	Saturday	10:00-12:59	14.0826	30.1376	9.4463	8.5236
7	Houston Denver	Tuesday	10:00-12:59	15.2464	41.2174	4.9430	4.3601
8	New York City South Florida	Thursday	07:00-09:59	16.3727	19.6367	3.3762	4.0214
9	North Florida Dallas	Sunday	Off-Peak (Late/Early)	16.0457	2.2837	2.2472	2.6316
10	Pittsburgh South Florida	Saturday	13:00-15:59	12.6357	23.7907	7.8125	6.6390
11	South Florida Boston	Sunday	10:00-12:59	15.1149	34.4015	4.8000	4.2179
12	South Florida Chicago	Sunday	10:00-12:59	17.9860	31.2573	3.0744	3.3520
13	South Florida DMV	Wednesday	13:00-15:59	14.2641	22.8327	9.8940	11.4790
14	South Florida Dallas	Thursday	10:00-12:59	14.4144	31.4457	2.6786	1.9100
15	South Florida Detroit	Tuesday	10:00-12:59	14.4980	35.7497	8.3851	8.8161
16	South Florida New York City	Monday	07:00-09:59	17.0013	21.7535	4.6338	4.5561
17	South Florida Pittsburgh	Saturday	10:00-12:59	12.9767	30.1634	8.0153	4.7619
18	South Florida St. Louis	Wednesday	10:00-12:59	13.7115	31.6079	3.6145	3.3101
19	St. Louis South Florida	Monday	13:00-15:59	14.6284	23.8151	4.4000	3.9312

1.7 Step 3 — Premium Extraction Classification

1.7.1 Why TOD drives classification (not DOW)

The DOW winner identifies *which day* to target. But pricing systems toggle premiums at the **time-of-day** level — not by individual calendar day. `tod_wup_share_pct` is therefore the primary classification signal.

1.7.2 Threshold derivation — data-driven, not arbitrary

All 20 corridor TOD winners were sorted by `tod_wup_share_pct` descending. Cut points were placed at **natural distributional gaps**, not round numbers:

Rank	Corridor	WUP%	Gap to next
5	Pittsburgh → South Florida	6.64%	↓ **1.88 pp gap**
6	South Florida → Pittsburgh	4.76%	
10	New York City → South Florida	4.02%	↓ **0.03 pp gap**
11	Chicago → South Florida	3.99%	
15	South Florida → Chicago	3.35%	↓ **0.04 pp gap**
16	South Florida → St. Louis	3.31%	

Cut points: > **6.5%** (top tier), **4.0%** (second tier), **3.32%** (third tier).

1.7.3 Category definitions

Category	Condition	Revenue action
Stronghold Extraction	wup > 6.5% AND velocity > 15%	Maximum premium +25–30% — dominant presence, protect and maximize
Surge Capture	4.0% wup 6.5% AND velocity > 15%	Aggressive push +20% — strong signal, room to grow share
Niche Protection	3.32% wup < 4.0%	Moderate uplift +10–15% — defend share against competitive displacement
Velocity Opportunity	velocity > 25% AND 2.0% wup < 3.32%	Selective push +15–20% — high market activity, build WUP concentration
No Signal	all other	No pricing action — insufficient evidence for extraction

1.7.4 Dual-criterion quality gate

Every active candidate satisfies **both** criteria: - Strong market demand: velocity 15% (or 25% for Velocity Opportunity) - Meaningful WUP presence: wup 2.0% across all categories (no sub-2% extractions)

18 of 20 corridors produce a `combined_rank = 2` winner — meaning the selected TOD slot ranks #1 on **both** velocity and WUP simultaneously. The remaining 2 are explained in the rank-sum section.

```
PREMIUM EXTRACTION CATEGORY DISTRIBUTION
------------------------------------------
category
Stronghold Extraction    5
Surge Capture            5
Niche Protection         5
Velocity Opportunity     3
No Signal                2
```

	corridor		dow_dimension	dow_velocity_pct
0	South Florida	Detroit	Tuesday	14.4980
1	DMV	South Florida	Friday	17.0175
2	Detroit	South Florida	Saturday	14.0826
3	Pittsburgh	South Florida	Saturday	12.6357
4	South Florida	DMV	Wednesday	14.2641
5	Houston	Denver	Tuesday	15.2464
6	South Florida	Boston	Sunday	15.1149
7	South Florida	Pittsburgh	Saturday	12.9767
8	South Florida	New York City	Monday	17.0013
9	New York City	South Florida	Thursday	16.3727
10	South Florida	Chicago	Sunday	17.9860
11	Dallas	Phoenix Valley	Sunday	15.0358
12	Dallas	South Florida	Wednesday	16.2885
13	St. Louis	South Florida	Monday	14.6284
14	Chicago	South Florida	Friday	16.3783
15	Dallas	Denver	Friday	15.8143
16	South Florida	St. Louis	Wednesday	13.7115
17	Chicago	Denver	Monday	14.5533
18	South Florida	Dallas	Thursday	14.4144
19	North Florida	Dallas	Sunday	16.0457

	dow_wup_share_pct	tod_dimension	tod_velocity_pct
0	8.3851	10:00-12:59	35.7497
1	9.4737	10:00-12:59	31.3032
2	9.4463	10:00-12:59	30.1376
3	7.8125	13:00-15:59	23.7907
4	9.8940	13:00-15:59	22.8327
5	4.9430	10:00-12:59	41.2174
6	4.8000	10:00-12:59	34.4015
7	8.0153	10:00-12:59	30.1634
8	4.6338	07:00-09:59	21.7535
9	3.3762	07:00-09:59	19.6367

```
10          3.0744           10:00-12:59          31.2573
11          2.7778           13:00-15:59          29.5943
12          3.0960           10:00-12:59          27.3828
13          4.4000           13:00-15:59          23.8151
14          3.3451           13:00-15:59          23.1257
15          3.7313           10:00-12:59          35.9166
16          3.6145           10:00-12:59          31.6079
17          3.2172           10:00-12:59          31.3305
18          2.6786           10:00-12:59          31.4457
19          2.2472  Off-Peak (Late/Early)          2.2837

    tod_wup_share_pct               category
0              8.8161  Stronghold Extraction
1             11.5880  Stronghold Extraction
2              8.5236  Stronghold Extraction
3              6.6390  Stronghold Extraction
4             11.4790  Stronghold Extraction
5              4.3601          Surge Capture
6              4.2179          Surge Capture
7              4.7619          Surge Capture
8              4.5561          Surge Capture
9              4.0214          Surge Capture
10             3.3520       Niche Protection
11             3.4274       Niche Protection
12             3.6832       Niche Protection
13             3.9312       Niche Protection
14             3.9900       Niche Protection
15             3.0668    Velocity Opportunity
16             3.3101    Velocity Opportunity
17             2.8643    Velocity Opportunity
18             1.9100              No Signal
19             2.6316              No Signal
```

1.8 Step 4 — Revenue Opportunity Quantification

1.8.1 Fleet parameters (Q1 Phenom 300)

Parameter	Low Scenario	High Scenario
Fleet hours (Q1)	3,400 hrs	6,800 hrs
Active tails	20	40
Hourly rate	$9,500	$9,500
Corridor allocation	20 corridors	20 corridors

1.8.2 SLI — Surge Loyalty Index

> **SLI** = `tod_wup_share_pct / NATIONAL_BASELINE` where `NATIONAL_BASELINE = 4.0%` (overall WUP Light Jet long-haul market share, from reference notebook Cell 5)

SLI measures how over- or under-indexed WUP is in the corridor's peak window **relative to the national average**: - **SLI = 1.0** → WUP presence equals the national average in that window - **SLI = 2.9** → WUP is 2.9× the national average (e.g. DMV → South Florida, 10:00–12:59) - **SLI < 1.0** → WUP is underweight; below market average in that window

---

1.8.3 Hours-in-Window Formula

> **`hours = (Fleet_hrs / N_corridors) × (tod_velocity_pct / 100) × SLI`**

Step-by-step:

Factor	What it does
`Fleet_hrs / N_corridors`	Pro-rata fleet allocation per corridor (170 hrs/corridor at low, 340 at high)
`× tod_velocity_pct / 100`	Scales to the fraction of those hours in the peak TOD window
`× SLI`	Adjusts for WUP's concentration — overweight corridors capture more revenue-eligible hours

---

1.8.4 Variable Toggle — Pricing Uplift within Category

The toggle is **not flat** within a category. It responds to signal strength:

Category	Condition for high toggle	High	Low
Stronghold Extraction	SLI > 2.0 (more than 2× national avg)	+30%	+25%
Surge Capture	—	+20%	+20%
Niche Protection	wup > 3.6%	+15%	+10%
Velocity Opportunity	velocity > 33%	+20%	+15%

Corridors with stronger signals within their tier receive the higher toggle — ensuring the pricing lever is proportional to conviction, not just category membership.

1.8.5 Value Formula

> **`value = hours_in_window × hourly_rate × toggle_pct`**

This is the **incremental revenue** from applying the pricing uplift during the peak extraction window. The model assumes the toggle is applied to WUP-bookable capacity — the SLI multiplier ensures we are only sizing the opportunity proportional to WUP's actual presence.

```
VALUE BY CATEGORY (Q1 Phenom Fleet)
------------------------------------------

                        value_low  value_high
category
Stronghold Extraction  $1,608,587  $3,217,697
Surge Capture            $521,930  $1,043,670
Niche Protection         $257,069    $514,282
Velocity Opportunity     $207,433    $415,008
No Signal                      $0          $0
```

```
                         corridor               category  \
0         South Florida   Detroit  Stronghold Extraction
1             DMV   South Florida  Stronghold Extraction
2         Detroit   South Florida  Stronghold Extraction
3      Pittsburgh   South Florida  Stronghold Extraction
4             South Florida   DMV  Stronghold Extraction
5                Houston   Denver          Surge Capture
6          South Florida   Boston          Surge Capture
7      South Florida   Pittsburgh          Surge Capture
8   South Florida   New York City          Surge Capture
9   New York City   South Florida          Surge Capture
10        South Florida   Chicago       Niche Protection
11        Dallas   Phoenix Valley       Niche Protection
12         Dallas   South Florida       Niche Protection
13      St. Louis   South Florida       Niche Protection
14        Chicago   South Florida       Niche Protection
15                Dallas   Denver   Velocity Opportunity
16      South Florida   St. Louis   Velocity Opportunity
17               Chicago   Denver   Velocity Opportunity
18         South Florida   Dallas              No Signal
19         North Florida   Dallas              No Signal

   tod_dimension  tod_velocity_pct  tod_wup_share_pct   sli  \
0    10:00-12:59           35.7497             8.8161  2.20
1    10:00-12:59           31.3032            11.5880  2.90
2    10:00-12:59           30.1376             8.5236  2.13
3    13:00-15:59           23.7907             6.6390  1.66
4    13:00-15:59           22.8327            11.4790  2.87
5    10:00-12:59           41.2174             4.3601  1.09
6    10:00-12:59           34.4015             4.2179  1.05
7    10:00-12:59           30.1634             4.7619  1.19
8    07:00-09:59           21.7535             4.5561  1.14
9    07:00-09:59           19.6367             4.0214  1.01
10   10:00-12:59           31.2573             3.3520  0.84
11   13:00-15:59           29.5943             3.4274  0.86
```

12	10:00-12:59	27.3828	3.6832	0.92
13	13:00-15:59	23.8151	3.9312	0.98
14	13:00-15:59	23.1257	3.9900	1.00
15	10:00-12:59	35.9166	3.0668	0.77
16	10:00-12:59	31.6079	3.3101	0.83
17	10:00-12:59	31.3305	2.8643	0.72
18	10:00-12:59	31.4457	1.9100	0.48
19	Off-Peak (Late/Early)	2.2837	2.6316	0.66

	toggle_pct	hours_low	hours_high	value_low	value_high
0	0.30	133.7	267.4	$381,045	$762,090
1	0.30	154.3	308.6	$439,755	$879,510
2	0.30	109.1	218.3	$310,935	$622,155
3	0.25	67.1	134.3	$159,362	$318,962
4	0.30	111.4	222.8	$317,490	$634,980
5	0.20	76.4	152.8	$145,160	$290,320
6	0.20	61.4	122.8	$116,660	$233,320
7	0.20	61.0	122.0	$115,900	$231,800
8	0.20	42.2	84.3	$80,180	$160,170
9	0.20	33.7	67.4	$64,030	$128,060
10	0.10	44.6	89.3	$42,370	$84,835
11	0.10	43.3	86.5	$41,135	$82,175
12	0.15	42.8	85.7	$60,990	$122,122
13	0.15	39.7	79.4	$56,572	$113,145
14	0.15	39.3	78.6	$56,002	$112,005
15	0.20	47.0	94.0	$89,300	$178,600
16	0.15	44.6	89.2	$63,555	$127,110
17	0.15	38.3	76.7	$54,578	$109,298
18	0.00	25.7	51.3	$0	$0
19	0.00	2.6	5.1	$0	$0

Active corridors : 18 / 20

	corridor	category	tod_dimension \
0	South Florida Detroit	Stronghold Extraction	10:00-12:59
1	DMV South Florida	Stronghold Extraction	10:00-12:59
2	Detroit South Florida	Stronghold Extraction	10:00-12:59
3	Pittsburgh South Florida	Stronghold Extraction	13:00-15:59
4	South Florida DMV	Stronghold Extraction	13:00-15:59
5	Houston Denver	Surge Capture	10:00-12:59
6	South Florida Boston	Surge Capture	10:00-12:59
7	South Florida Pittsburgh	Surge Capture	10:00-12:59
8	South Florida New York City	Surge Capture	07:00-09:59
9	New York City South Florida	Surge Capture	07:00-09:59
10	South Florida Chicago	Niche Protection	10:00-12:59
11	Dallas Phoenix Valley	Niche Protection	13:00-15:59
12	Dallas South Florida	Niche Protection	10:00-12:59
13	St. Louis South Florida	Niche Protection	13:00-15:59
14	Chicago South Florida	Niche Protection	13:00-15:59

```
15              Dallas   Denver   Velocity Opportunity   10:00-12:59
16       South Florida  St. Louis   Velocity Opportunity   10:00-12:59
17             Chicago   Denver   Velocity Opportunity   10:00-12:59

   tod_velocity_pct  tod_wup_share_pct   sli  toggle_pct  hours_low  \
0           35.7497             8.8161  2.20        0.30      133.7
1           31.3032            11.5880  2.90        0.30      154.3
2           30.1376             8.5236  2.13        0.30      109.1
3           23.7907             6.6390  1.66        0.25       67.1
4           22.8327            11.4790  2.87        0.30      111.4
5           41.2174             4.3601  1.09        0.20       76.4
6           34.4015             4.2179  1.05        0.20       61.4
7           30.1634             4.7619  1.19        0.20       61.0
8           21.7535             4.5561  1.14        0.20       42.2
9           19.6367             4.0214  1.01        0.20       33.7
10          31.2573             3.3520  0.84        0.10       44.6
11          29.5943             3.4274  0.86        0.10       43.3
12          27.3828             3.6832  0.92        0.15       42.8
13          23.8151             3.9312  0.98        0.15       39.7
14          23.1257             3.9900  1.00        0.15       39.3
15          35.9166             3.0668  0.77        0.20       47.0
16          31.6079             3.3101  0.83        0.15       44.6
17          31.3305             2.8643  0.72        0.15       38.3

   hours_high value_low value_high
0       267.4  $381,045   $762,090
1       308.6  $439,755   $879,510
2       218.3  $310,935   $622,155
3       134.3  $159,362   $318,962
4       222.8  $317,490   $634,980
5       152.8  $145,160   $290,320
6       122.8  $116,660   $233,320
7       122.0  $115,900   $231,800
8        84.3   $80,180   $160,170
9        67.4   $64,030   $128,060
10       89.3   $42,370    $84,835
11       86.5   $41,135    $82,175
12       85.7   $60,990   $122,122
13       79.4   $56,572   $113,145
14       78.6   $56,002   $112,005
15       94.0   $89,300   $178,600
16       89.2   $63,555   $127,110
17       76.7   $54,578   $109,298
```

```
<IPython.core.display.HTML object>
```

1.9 Results Interpretation Guide

1.9.1 Priority order for revenue management action

Priority	Category	Action
1	**Stronghold Extraction**	Implement maximum toggles immediately — high conviction, high share
2	**Surge Capture**	Deploy within the week — strong signal, competitive windows
3	**Niche Protection**	Schedule for next cycle — moderate, protect before competition moves
4	**Velocity Opportunity**	Monitor and test — demand is real but WUP share is still building

1.9.2 Reading the SLI column

SLI is the clearest single indicator of how well-positioned WUP is in a corridor-window pair. Any SLI > 1.5 represents a meaningfully above-average WUP concentration — a reliable extraction signal. SLI > 2.0 (Stronghold threshold) represents a dominant WUP position that should be monetized aggressively.

1.9.3 Low vs High scenario

The low/high range reflects Q1 fleet deployment uncertainty (20 vs 40 tails). Both scenarios use identical toggle logic — the range scales linearly with fleet hours. Use low as a floor for planning; high as upside for investment cases.